

Predicting When Collection Statistics Help Top- k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

Aviral Gupta Mohit Kumar Singh Anirudh Sankar Arihant Tripathy

{aviral.gupta, mohit.singh, anirudh.sankar, arihant.tripathy}@research.iiit.ac.in
IIIT Hyderabad, India

Abstract

On sparse-autoencoder features, BM25 scores 0.582 nDCG@10 where a plain inner product over the same vectors scores 0.381. On SPLADE the ordering reverses, 0.455 against 0.568. What BM25 adds is collection frequency, the rate at which a feature fires across a corpus, and the published account of the reversal is that this helps only while a vocabulary stays Zipfian. That account has been observed one vocabulary at a time and never made predictive, so for a new sparse representation the only way to pick a scorer is to run both. We make that choice predictable. Eight vocabularies built from one frozen encoder are scored with a BM25-style function whose four components toggle independently, and we regress the gap between its corrected and uncorrected forms on six distributional measurements that cost one forward pass, with one row per vocabulary and collection, since a vocabulary has no single frequency curve. Block-sparse attention and mixture-of-experts routing pose the same choice and consult no such statistic either, so the fitted predictor goes to them without refitting, in units that carry across the three settings. Whether it holds there is an open question. Attention sinks fire for almost every query yet cannot be removed, and the rarest experts are the most critical, so frequency may signal importance differently inside a model than across a corpus. Evaluation covers fourteen collections, four of them Indic or code-mixed, where the distributions the correction depends on are known to shift.

1 Motivation and Research Gap

A learned sparse retriever scores a query-document pair as $\langle u_q, u_d \rangle$ over a concept vocabulary. A block-sparse attention kernel estimates how critical KV page i is for query q_t as $r_{t,i} = \langle q_t, c_i \rangle$ over page summaries, then attends to the top- k pages only (Tang et al., 2024). A mixture-of-experts router gates on $g_e = \langle h_t, w_e \rangle$ and sends the token to its

top- k experts (Shazeer et al., 2017; Fedus et al., 2022). These belong to three different literatures and perform one computation, an inner product against a learned non-negative basis followed by a hard top- k .

What all three leave out is any account of how often a given feature wins. Classical retrieval settled that question decades ago, and the correction has three parts, namely inverse document frequency, which discounts features that fire everywhere, term-frequency saturation, so that repeated evidence stops accumulating linearly, and length normalisation (Robertson and Zaragoza, 2009). Parts of it have been reinvented inside the transformer under other names. Loss-free load balancing adds a per-expert bias driven by recent load (Wang et al., 2024), an additive frequency discount applied during training with no saturation or length term, and the auxiliary losses it replaces push toward uniform expert use (Zoph et al., 2022). Streaming inference hand-codes an exception for the highest-frequency keys in the cache (Xiao et al., 2024). Each of these is a local fix.

The shortcoming in the published work is narrower than those local fixes suggest. Two papers have run the comparison, both in retrieval, and both explain their result the same way, that BM25 helps while a vocabulary keeps its Zipfian structure (Clavié et al., 2026; Han et al., 2026). Neither varies the vocabulary systematically, so the explanation rests on one vocabulary at a time. It has never been turned into a quantity computable before the experiment, and never checked where the same computation runs inside a model.

2 Contributions

1. A controlled family of sixteen representations from one frozen encoder, so basis construction and granularity vary while representation learning stays fixed. The published comparisons vary

one vocabulary at a time.

2. One scoring function with four independent switches, so the plain inner product and BM25 are two configurations of the same function and any gain names a single mechanism.
3. A predictor in place of an explanation. Six diagnostics are regressed on the gap between those two configurations over 224 rows and validated on held-out basis families and held-out collections, which turns the Zipfian account into something usable in advance.
4. A test outside retrieval, which no single-setting study can run. The fitted predictor is applied without refitting to KV-page selection and expert routing, in units that carry across the three settings.

3 Background

BM25 scores a document d against query Q as

$$\sum_{i \in Q} \text{idf}_i \cdot \frac{f_i(d) (k_1 + 1)}{f_i(d) + k_1 \left(1 - b + b \frac{|d|}{dl}\right)}, \quad (1)$$

with $\text{idf}_i = \ln\left(1 + \frac{N - df_i + 0.5}{df_i + 0.5}\right)$. None of these ingredients requires words. Each needs only a representation with non-negative coordinates, a meaningful per-dimension collection frequency df_i , and a magnitude that admits a notion of length. Dense embeddings have none of these, since their coordinates are signed and every dimension is active in every document, which is why dense retrieval is scored with a plain inner product.

Learned sparse features change the picture. Sparse-autoencoder activations, k -means codes and quantised codebooks are non-negative, sparse and count-like. KV page summaries are built from signed keys and so are not non-negative themselves, but the quantity IDF needs is a selection count, which is non-negative for pages and experts alike and has a well-defined rate over a calibration set. All three mechanisms are eligible for the same correction, so whether it helps is an empirical question.

4 Related Work

Learned sparse retrieval and its vocabularies. SPLADE and its successors learn term weights over the backbone subword vocabulary and retain the inverted index (Formal et al., 2021; Lassance et al., 2024). Recent work replaces that vocabulary with a learned one. CL-SR indexes SAE-decomposed dense embeddings as concept units

(Park et al., 2025), SPLARE trains an SAE-based sparse retriever directly (Formal et al., 2026), and SAE-SPLADE swaps SPLADE’s MLM head for an SAE encoder (Zong et al., 2026). All three score with an inner product.

Scoring learned sparse features with BM25. Clavié et al. (2026) show that SAE features taken from a frozen dense retriever have roughly Zipfian collection statistics and are directly BM25-scorable. Their Appendix C holds the sharpest evidence for our question. On one suite BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms, and 0.455 against 0.568 on SPLADE-v3, so the two scorers swap places once the vocabulary changes beneath them. Han et al. (2026) run the comparison on image features and find (their §4.4) that the two differ by at most about two points of R@1, attributing the parity to top- k filtering already suppressing what IDF would down-weight. The same section reports that raising the per-patch activation budget from $k=16$ to $k=128$ makes document frequencies converge and drops R@1 from 0.508 to 0.038, so the effect is distributional. Both converge on the Zipfian explanation of Section 1, which is in print already and not ours to supply.

Selection inside the transformer. Quest scores KV pages by an inner product against page min-max key summaries and loads only the top- k (Tang et al., 2024). H2O keeps a mix of recent and heavy-hitter tokens (Zhang et al., 2023b), and StreamingLLM preserves the earliest keys as attention sinks (Xiao et al., 2024). Expected Attention ranks a cached pair by how strongly future queries are predicted to attend to it and prunes the low scorers (Devoto et al., 2025). None of these uses the rate at which a page is selected across queries. On the routing side, Su et al. (2026) show that a handful of rare experts are disproportionately critical. To our knowledge no work treats retrieval scoring, KV selection and routing as one scoring problem.

5 Proposed Methodology

A modular scorer. For every setting we implement one scoring function in which soft IDF weighting, term-frequency saturation, length normalisation and top- k sparsification are separate switches. The uncorrected inner product is the all-off configuration and BM25 is the all-on configuration. The quantity we predict throughout is the

margin,

$$m = \frac{Q_{\text{on}} - Q_{\text{off}}}{s(S, c)}, \quad (2)$$

the change in quality between two configurations at matched budget, divided by the standard deviation $s(S, c)$ of that quality across the fixed set of switch configurations in setting S on collection c . The margin is then dimensionless and means the same thing in every setting, which is what lets a model fitted in one be scored in another. The primary metric is nDCG@10 in Setting A, RULER accuracy in Setting B and MMLU accuracy in Setting C, with perplexity entering as its negative so that every metric points the same way.

Setting A: retrieval, used for calibration. We build eight concept bases from a single frozen encoder, spanning near-dense to strongly lexical, namely a split-rectified identity map, random projection, PCA, k -means codes, product quantisation (Jégou et al., 2011), residual quantisation, a Top- K sparse autoencoder (Gao et al., 2025), and SPLADE (Formal et al., 2021; Lassance et al., 2024). SPLADE is in the set on purpose, since the reversal we explain is SPLADE against SAE. Each basis is instantiated at pooled and token-level granularity, giving sixteen representations, with granularity as an explicit variable.

Setting B: KV-page selection. We start from Quest-style page criticality (Tang et al., 2024) and add an online statistic sf_i , the fraction of prefill queries for which page i enters the top- k . The corrected score multiplies the saturated criticality by $\text{idf}(\text{sf}_i)$ and by a length term in $\|c_i\|$, which is the scorer of Eq. (1) with pages for documents. The method needs no training and drops into an existing kernel. Whether attention sinks are exempted from the correction is an ablation rather than an assumption. If IDF suppresses them and accuracy falls, that outcome is the mechanism result.

Setting C: expert routing. The same correction applies to router logits at inference time on OLMoE-1B-7B (Muennighoff et al., 2025), which activates 8 of 64 experts per layer, with expert selection frequency measured over calibration data as in Setting B. Loss-free balancing (Wang et al., 2024) is the additive, training-time special case we compare against, and we check whether the correction protects or destroys the Super Experts of Su et al. (2026).

Diagnostics and the predictor. From a single forward pass over calibration data we compute six measurements, namely active fraction per item, the Zipf exponent of the selection-frequency curve, the Gini coefficient of activation mass, head mass, the variance of IDF values, and sink share, meaning the fraction of score mass held by the top 1% most-frequently-selected features. Han et al. (2026) report several descriptively, with power-law exponents in [1.20, 2.32], but predict nothing with them. One row of the regression pairs a representation with a collection, because collection frequency belongs to a corpus and a basis therefore has no single frequency curve. Measuring it again on each collection is free, since indexing already performs that pass. Sixteen representations across the fourteen collections of Section 6 give 224 rows for six regressors.

Theory. We aim to characterise when an IDF-weighted linear scorer over a learned basis provably reduces to a bilinear inner product on the underlying dense embeddings, which we call the collapse condition. Where it holds the two scorers must agree, which explains part of the grid instead of tabulating it.

6 Datasets

Setting A runs over fourteen collections. Bases are fitted and hyperparameters tuned on MS MARCO passage (Bajaj et al., 2016), which supplies 8.8M passages, and tuning happens nowhere else, so every other collection tests generalisation. LIMIT (Weller et al., 2025) separates lexical from semantic signal by construction and should make any BM25 advantage easy to see. TREC-DL 19 and 20 and six BEIR datasets (Thakur et al., 2021), namely NFCorpus, SciFact, FiQA-2018, TREC-COVID, ArguAna and SCIDOCs, cover the domains over which the published reversal was reported.

Because collection frequency is a property of a language and a corpus rather than of a model, the remaining four collections are the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a) and a set of Hindi-English code-mixed queries, where transliteration and mixing distort the document-frequency curve. Neither prior SAE-based sparse retriever touches an Indic language, and the multilingual evaluation in Zong et al. (2026) covers six MIRACL languages of which none is Indian. Corpus-level measures of mixing already predict downstream behaviour there (Kodali et al.,

2022). Because every collection contributes its own rows, these four enter the fit itself, and they add under 2M documents.

Setting B uses RULER (Hsieh et al., 2024), needle-in-a-haystack and LongBench on Llama-3.2-1B/3B-Instruct and Qwen2.5-7B-Instruct. Setting C uses OLMoE-1B-7B with WikiText for perplexity and MMLU and GSM8K for downstream accuracy.

7 Evaluation Plan

For Setting A we report nDCG@10, MRR@10 and Recall@100/1000 alongside index size, FLOPs and latency, since a gain that destroys the efficiency of sparsity is worth less. As a pipeline check we reproduce that Appendix C reversal within a stated tolerance before trusting our own numbers. Setting B reports accuracy at matched token budget against Quest and H2O (Tang et al., 2024; Zhang et al., 2023b), and Setting C reports MMLU, GSM8K, perplexity and routing entropy.

The headline result is the diagnostics-to-margin regression. The fitted model emits three quantities we can score, the sign of the margin, measured by accuracy and AUC, its magnitude in units of s , measured by held-out R^2 within retrieval and by mean absolute error when the retrieval-fitted model is carried to Settings B and C without refitting, and the ordering of configurations within a setting, measured by Kendall’s τ . Rows are nested by representation and by collection, so we cluster standard errors on both axes and validate by holding out whole groups on each, four basis families and fourteen collections, reported separately, since generalising to an unseen construction method and generalising to an unseen corpus are different claims. Much of the raw variance sits between collections, so the number we report is the gain in held-out R^2 over a baseline carrying collection indicators alone, so no model can score well by learning which corpus is easy.

The analysis is fixed in advance, since the grid produces several hundred pairwise comparisons. The primary comparison is all-on against all-off per representation and component ablations are secondary. Significance uses a paired bootstrap over queries with Holm–Bonferroni correction within each family of comparisons, effect sizes are reported alongside p -values, and rows are weighted by the inverse variance of the margin estimate, since a small code-mixed query set gives a noisier

margin than MS MARCO dev. The transfer holds if the predicted sign is correct on at least 70% of held-out rows in Settings B and C and the mean absolute error there is no worse than 1.5 times the residual spread inside retrieval.

8 Feasibility

Nine weeks separate this proposal from the final submission. The retrieval grid over all fourteen collections runs first and Setting B follows it, so a transformer result exists by the mid submission, leaving the predictor, the cross-setting transfer and the ablations for the weeks after that. The four of us split the work by setting, with the scorer and the diagnostics shared.

The hardware available to us provides nodes of four GPUs with 11 GB of device memory each, which shapes the design. MS MARCO is encoded once with a frozen DistilBERT-scale encoder and every pooled basis is a linear or quantisation map over the cached embeddings, with token-level granularity capped at a 1M-passage subsample. Settings B and C are inference-only, and we chose OLMoE-1B-7B over a checkpoint such as Qwen3-30B-A3B because its 6.9B parameters fit a node at 4-bit where the 30B model would not, while still exposing 64 experts per layer.

9 Risks and Limitations

Confirming that BM25 helps on Zipfian vocabularies would only repeat what the cited papers predict, so the informative outcomes are the awkward ones. A predicted gain where the measured effect is negative would show contribution to the output and discrimination between queries coming apart inside the transformer. Diagnostics promising a gain that never appears would falsify the Zipfian account as stated. If the predictor is weak on both held-out axes we report the grid as a benchmark and the collapse condition as the theoretical contribution, a negative result with a resource attached. If Setting C proves infeasible we fall back to Settings A and B, which still support the transfer claim.

Three limitations stand out. Setting A uses a single frozen encoder, so basis and encoder effects are not fully separable. Setting C is inference-time only, so we claim nothing about training dynamics. The diagnostics are correlational, and only the transfer test is causal.

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